

Spatial Transcriptomics Cheatsheet

The tables below consist of valuable functions or commands that will help you through this module.

Each table represents a different library/tool and its corresponding commands.

Please note that these tables are not intended to tell you all the information you need to know about each command.

The hyperlinks found in each piece of code will take you to the documentation for further information on the usage of each command. Please be aware that the documentation will generally provide information about the given function's most current version (or a recent version, depending on how often the documentation site is updated). This will usually (but not always!) match what you have installed on your machine. If you have a different version of R or other R packages, the documentation may differ from what you have installed.

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SpatialExperiment

Read the [SpatialExperiment package documentation \(and e-book\)](#) and a [vignette on its usage](#).

`SpatialExperiment` objects are based on `SingleCellExperiment` objects. There are many functions that are shared between the two, with equivalent usage. See [this SingleCellExperiment vignette](#) for example usage of shared functions. You can also refer to the Data Lab cheatsheets for our "Introduction to scRNA-seq" (`scRNA-seq-cheatsheet.pdf`) and "Advanced scRNA-seq" (`scRNA-seq-advanced-cheatsheet.pdf`) training modules.

Library/Package	Piece of Code	What it's called	What it does
<code>SpatialExperiment</code>	<code>SpatialExperiment()</code>	Spatial Experiment	Creates a <code>SpatialExperiment</code> object
<code>SpatialExperiment</code>	<code>spatialCoords()</code>	Spatial Experiment coordinates	Extracts spatial coordinates of spots in the <code>SpatialExperiment</code> object
<code>SpatialExperiment</code>	<code>imgData()</code>	Spatial Experiment images	View the image data associated with a <code>SpatialExperiment</code> object

VisiumIO

Read the R package [VisiumIO documentation](#) and a [vignette on its usage](#).

Library/Package	Piece of Code	What it's called	What it does
<code>VisiumIO</code>	<code>TENxVisium()</code>	10x Visium	Constructor function for importing 10x Visium data
<code>VisiumIO</code>	<code>import()</code>	Import	Loads 10x Visium data defined with the <code>TENxVisium()</code> constructor function

ggspavis

Read the [ggspavis package documentation](#) and a [vignette on its usage](#).

Library/Package	Piece of Code	What it's called	What it does
<code>ggspavis</code>	<code>plotCoords()</code>	Plot coordinates	Creates a spot plot showing spatial locations in x/y with options for spot annotation
<code>ggspavis</code>	<code>plotVisium()</code>	Plot Visium	Creates a spot plot for Visium data specifically showing spatial locations in x/y with options for spot annotation as well as an option to show the spatial image, e.g., the H&E image

SpotSweeper

Read the [SpotSweeper package documentation](#) and a [vignette on its usage](#).

Library/Package	Piece of Code	What it's called	What it does
SpotSweeper	<code>localOutliers()</code>	Local outliers	Detect local outlier spots in a <code>SpatialExperiment</code> based on a given quality control metric
SpotSweeper	<code>plotQCMetrics()</code>	Plot QC metrics	Create a spot plot highlighting outliers detected with <code>SpotSweeper::localOutliers()</code>

Banksy

Read the [Banksy package documentation](#), which contains several vignettes on its usage, including a [vignette on parameter selection](#). Note that many of their vignettes use of image-based spatial transcriptomics data that is more fine-grained compared to 10x Visium.

Library/Package	Piece of Code	What it's called	What it does
Banksy	<code>computeBanksy()</code>	Compute <code>Banksy</code>	Compute the component neighborhood matrices for the <code>Banksy</code> matrix
Banksy	<code>runBanksyPCA()</code>	Run <code>Banksy</code> PCA	Run PCA on a <code>Banksy</code> matrix with a weighted spatial neighborhood context

spacexr

Read the [spacexr package documentation](#), and a [vignette on using their RCTD algorithm](#).

Library/Package	Piece of Code	What it's called	What it does
spacexr	<code>createRctd()</code>	Create an RCTD object	Preprocess data before performing deconvolution with RCTD
spacexr	<code>runRCTD()</code>	Run RCTD	Run the RCTD algorithm to decompose cell type mixtures
spacexr	<code>plotCellTypeWeight()</code>	Plot a single cell type's weights	Plot pixel proportions for a specific cell type on a spatial coordinates plot

scuttle, scran, and scater

Read the [scuttle package documentation](#), and a [vignette on its usage](#).

Read the [scran package documentation](#), and a [vignette on its usage](#).

Read the [scater package documentation](#), and a [vignette on its usage](#).

Library/Package	Piece of Code	What it's called	What it does
scuttle	<code>addPerCellQC()</code>	Add per cell quality control	For a <code>SingleCellExperiment</code> object, calculate and add quality control per cell and store in <code>colData</code>
scuttle	<code>computeLibraryFactors()</code>	Compute Library Factors	Returns a numeric vector of computed size factors for each spot (or cell) stored in a <code>SpatialExperiment</code> (or <code>SingleCellExperiment</code>) object. The size factor is computed as the library size of each spot/cell after scaling them to have a mean of 1 across all spots/cells
scuttle	<code>logNormCounts()</code>	Normalize log counts	Returns the <code>SpatialExperiment</code> (or <code>SingleCellExperiment</code>) object with normalized expression values for each spot (cell), using the size factors stored in the object
scran	<code>getTopHVGs()</code>	Get top highly variable genes	Identify variable genes in a <code>SingleCellExperiment</code> object, based on variance
scran	<code>modelGeneVar()</code>	model per gene variance	Model the per gene variance of a <code>SingleCellExperiment</code> object
scran	<code>clusterCells()</code>	Cluster cells	Perform clustering on an SCE object using the <code>bluster</code> package
scater	<code>runPCA()</code>	Run PCA	Calculates principal components analysis on a <code>SingleCellExperiment</code> object, returning an SCE object with a PCA reduced dimension
scater	<code>runUMAP()</code>	Run UMAP	Calculates uniform manifold approximate projection on a <code>SingleCellExperiment</code> object, returning an SCE object with a UMAP reduced dimension
scater	<code>plotUMAP()</code>	Plot UMAP	Plot the "UMAP"-named reduced dimension slot from a <code>SingleCellExperiment</code> object

bluster

Read the [bluster package documentation](#) and vignettes on its usage:

- [Flexible clustering for Bioconductor](#)
- [Assorted clustering diagnostics](#)

Library/Package	Piece of Code	What it's called	What it does
bluster	<code>NNGraphParam()</code>	Graph-based clustering parameters	Set up parameters for nearest-neighbor (NN) graph-based clustering algorithms within <code>scrn::clusterCells()</code> or <code>bluster::clusterRows()</code>
bluster	<code>approxSilhouette()</code>	Approximate silhouette width	Calculate an approximate silhouette width for each cell given a set of clusters
bluster	<code>neighborPurity()</code>	Compute neighborhood purity	Calculate neighborhood purity for each cell given a set of clusters
bluster	<code>bootstrapStability()</code>	Assess cluster stability by bootstrapping	Generate cluster bootstrap replicates to estimate cluster robustness to sampling noise
bluster	<code>pairwiseRand()</code>	Compute pairwise Rand indices	Calculate the Rand index (adjusted, by default) between pairs of clustering results, using one of several available calculation modes

patchwork

Read the [patchwork package documentation](#), and a [vignette on its usage](#).

Library/Package	Piece of Code	What it's called	What it does
patchwork	<code>wrap_plots()</code>	Wrap plots	Wrap multiple <code>ggplot2</code> objects into a single multi-panel plot
patchwork	<code>+</code>	Plot arithmetic	Place <code>ggplot2</code> objects side-by-side by adding them together with a <code>+</code> . The <code>patchwork</code> package must be loaded to use this symbol with plots
patchwork	<code>/</code>	Plot arithmetic	Stack <code>ggplot2</code> objects on top of one another "dividing" them with a <code>/</code> . The <code>patchwork</code> package must be loaded to use this symbol with plots

pals

Read the [pals package documentation](#), and a [vignette on its usage](#).

Library/Package	Piece of Code	What it's called	What it does
pals	<code>alphabet()</code>	Alphabet palette	Create a discrete palette from the "alphabet" set of colors

pheatmap

Read the [pheatmap package documentation](#).

Library/Package	Piece of Code	What it's called	What it does
pheatmap	<code>pheatmap()</code>	Pretty heatmap	Plot a (pretty!) clustered heatmap

purrr

Read the [purrr package documentation](#) and a [vignette on its usage](#), and download the [purrr package cheatsheet](#).

Library/Package	Piece of Code	What it's called	What it does
purrr	<code>map()</code>	map	Apply a function across each element of list; return a list
purrr	<code>imap()</code>	imap	Apply a function across each element of list and its index/names; return a list
purrr	<code>reduce()</code>	Reduce	Reduce a list to a single value by repeatedly applying a given function. Can also be used to iteratively modify a single object.